

# Basis Data Part4

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PENGAMPU:

KHAMARUDIN SYARIF

# Fungsi Skalar Matematika

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| Fungsi                 | Deskripsi                                                             | Contoh                         | Hasil      |
|------------------------|-----------------------------------------------------------------------|--------------------------------|------------|
| <code>abs()</code>     | Mengembalikan nilai absolute dari nilai input                         | <code>abs(-17.4)</code>        | 17.4       |
| <code>ceiling()</code> | Mengembalikan nilai integer terbesar yang terdekat dengan nilai input | <code>ceiling(-95.3)</code>    | -95        |
| <code>floor()</code>   | Mengembalikan nilai integer terkecil yang terdekat dengan nilai input | <code>floor(-42.8)</code>      | -43        |
| <code>round()</code>   | Mengembalikan nilai pembulatan dari suatu nilai decimal               | <code>round(42.4382, 2)</code> | 42.44      |
| <code>sqrt()</code>    | Mengembalikan nilai akar kuadrat dari nilai input                     | <code>sqrt(2.0)</code>         | 1.41421356 |
| <code>mod()</code>     | Mengembalikan nilai sisa hasil pembagian dari nilai input             | <code>mod(9,4)</code>          | 1          |
| <code>exp()</code>     | Mngembalikan nilai ekponensial dari nilai input                       | <code>exp(1.0)</code>          | 2.71828183 |

# Fungsi Skalar – ABS()

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```
1 SELECT StudentID, FirstName, LastName, Semester1, Semester2, ABS(MarkGrowth)
   as MarkGrowth
2 FROM students;
```

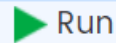
| StudentID | FirstName | LastName | Semester1 | Semester2 | MarkGrowth |
|-----------|-----------|----------|-----------|-----------|------------|
| 1         | Jose      | Mohit    | 64.55     | 72.60     | 8.05       |
| 2         | Lala      | Karlina  | 72.85     | 65.35     | 7.50       |
| 3         | Sultan    | Hadi     | 45.32     | 50.25     | 4.93       |
| 4         | Jaya      | Usman    | 86.73     | 77.40     | 9.33       |
| 5         | Anjali    | Wijaya   | 92.25     | 90.75     | 1.50       |



## Fungsi Skalar – CEILING()

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```
1 SELECT StudentID, FirstName, LastName, CEILING(Semester1) as Semester1,  
   CEILING(Semester2) as Semester2, MarkGrowth  
2 FROM students;
```

| StudentID | FirstName | LastName | Semester1 | Semester2 | MarkGrowth |
|-----------|-----------|----------|-----------|-----------|------------|
| 1         | Jose      | Mohit    | 65        | 73        | -8.05      |
| 2         | Lala      | Karlina  | 73        | 66        | 7.50       |
| 3         | Sultan    | Hadi     | 46        | 51        | -4.93      |
| 4         | Jaya      | Usman    | 87        | 78        | 9.33       |
| 5         | Anjali    | Wijaya   | 93        | 91        | 1.50       |

# Fungsi Skalar – FLOOR()

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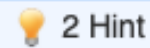
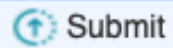
```
1 SELECT StudentID, FirstName, LastName, FLOOR(Semester1) as Semester1,  
   FLOOR(Semester2) as Semester2, MarkGrowth  
2 FROM students;
```

| StudentID | FirstName | LastName | Semester1 | Semester2 | MarkGrowth |
|-----------|-----------|----------|-----------|-----------|------------|
| 1         | Jose      | Mohit    | 64        | 72        | -8.05      |
| 2         | Lala      | Karlina  | 72        | 65        | 7.50       |
| 3         | Sultan    | Hadi     | 45        | 50        | -4.93      |
| 4         | Jaya      | Usman    | 86        | 77        | 9.33       |
| 5         | Anjali    | Wijaya   | 92        | 90        | 1.50       |



## Fungsi Skalar – ROUND()

Code Editor



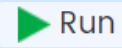
```
1 SELECT StudentID, FirstName, LastName, ROUND(Semester1,1) as Semester1,  
   ROUND(Semester2,0) as Semester2, MarkGrowth  
2 FROM students;
```

| StudentID | FirstName | LastName | Semester1 | Semester2 | MarkGrowth |
|-----------|-----------|----------|-----------|-----------|------------|
| 1         | Jose      | Mohit    | 64.6      | 73        | -8.05      |
| 2         | Lala      | Karlina  | 72.9      | 65        | 7.50       |
| 3         | Sultan    | Hadi     | 45.3      | 50        | -4.93      |
| 4         | Jaya      | Usman    | 86.7      | 77        | 9.33       |
| 5         | Anjali    | Wijaya   | 92.3      | 91        | 1.50       |

# Fungsi Skalar – SQRT()

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Code Editor



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0 Hint



```
1 SELECT StudentID, FirstName, LastName, SQRT(Semester1) as Semester1,  
   Semester2, MarkGrowth  
2 FROM students;
```

| StudentID | FirstName | LastName | Semester1  | Semester2 | MarkGrowth |
|-----------|-----------|----------|------------|-----------|------------|
| 1         | Jose      | Mohit    | 8.03430146 | 72.60     | -8.05      |
| 2         | Lala      | Karlina  | 8.53522115 | 65.35     | 7.50       |
| 3         | Sultan    | Hadi     | 6.73201307 | 50.25     | -4.93      |
| 4         | Jaya      | Usman    | 9.31289429 | 77.40     | 9.33       |
| 5         | Anjali    | Wijaya   | 9.60468636 | 90.75     | 1.50       |

# Quiz 1:

Gunakan fungsi **MOD()** untuk menghitung nilai sisa jika nilai Semester1 dibagi 2 dan fungsi **EXP()** untuk menghitung nilai eksponensial dari nilai MarkGrowth. Gunakan kedua fungsi tersebut dalam satu SELECT-Statement.

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| Console   |           | Graphic  |                    |           |                        |  |
|-----------|-----------|----------|--------------------|-----------|------------------------|--|
| StudentID | FirstName | LastName | Semester1          | Semester2 | EXP(MarkGrowth)        |  |
| 1         | Jose      | Mohit    | 0.5499999999999972 | 72.6      | 0.00031910192248120326 |  |
| 2         | Lala      | Karlina  | 0.8499999999999943 | 65.35     | 1808.0424144560632     |  |
| 3         | Sultan    | Hadi     | 1.3200000000000003 | 50.25     | 0.0072265032813764625  |  |
| 4         | Jaya      | Usman    | 0.7300000000000004 | 77.4      | 11271.131485524471     |  |
| 5         | Anjali    | Wijaya   | 0.25               | 90.75     | 4.4816890703380645     |  |



# Fungsi Text

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| Fungsi                    | Deskripsi                                                                                                                                                                | Contoh                                                                                                                                        | Hasil                   |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <code>concat()</code>     | Menggabungkan semua argumen/input, NULL value akan diabaikan                                                                                                             | <code>concat('abcde', 2, NULL, 22)</code>                                                                                                     | abcde222                |
| <code>split_part()</code> | Membagi string/text berdasarkan delimiter/pemisah yang ditentukan, kemudian mengembalikan value berdasarkan posisi string yang diinginkan (hitungan posisi mulai dari 1) | <code>split_part('abc@def@ghi', '@', 2)</code>                                                                                                | def                     |
| <code>substr()</code>     | Mengekstrak karakter/string yang diinginkan                                                                                                                              | <code>substr('alphabet', 3, 2)</code>                                                                                                         | ph                      |
| <code>length()</code>     | Menghitung jumlah karakter dalam string/text                                                                                                                             | <code>length('jose')</code>                                                                                                                   | 4                       |
| <code>replace()</code>    | Mengganti karakter dalam suatu string/text                                                                                                                               | <code>replace('abcdefabcdef', 'cd', 'XX')</code>                                                                                              | abXXefabXXef            |
| <code>trim()</code>       | Menghapus karakter dalam suatu string                                                                                                                                    | <code>trim(both 'x' from 'xTomxx');</code><br><code>trim(leading 'x' from 'xTomxx');</code><br><code>trim(trailing 'x' from 'xTomxx');</code> | Tom;<br>Tomxx;<br>xTom; |
| <code>upper()</code>      | Mengubah huruf kecil ke huruf besar                                                                                                                                      | <code>upper('tom')</code>                                                                                                                     | TOM                     |
| <code>lower()</code>      | Mengubah huruf besar ke huruf kecil                                                                                                                                      | <code>lower('TOM')</code>                                                                                                                     | tom                     |

# Fungsi Text – CONCAT()

---

```
Code Editor ▶ Run ↻ Submit 💡 0 Hint ⋮  
1 SELECT StudentID, CONCAT(FirstName, LastName) as Name, Semester1, Semester2,  
   MarkGrowth  
2 FROM students;
```

| StudentID | Name         | Semester1 | Semester2 | MarkGrowth |
|-----------|--------------|-----------|-----------|------------|
| 1         | JoseMohit    | 64.55     | 72.60     | -8.05      |
| 2         | LalaKarlina  | 72.85     | 65.35     | 7.50       |
| 3         | SultanHadi   | 45.32     | 50.25     | -4.93      |
| 4         | JayaUsman    | 86.73     | 77.40     | 9.33       |
| 5         | AnjaliWijaya | 92.25     | 90.75     | 1.50       |

# Fungsi Text – SUBSTRING\_INDEX()

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Syntax:

```
SELECT SUBSTRING_INDEX(column, delimiter, index to return)  
FROM TableName;
```

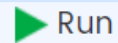
Keterangan:

- column* --> merupakan nama kolom yang akan dipecah text-nya,
- delimiter* --> karakter atau gabungan beberapa karakter untuk memecah text pada kolom bersangkutan,
- index\_to\_return* --> indeks dari pecahan text yang akan diambil.

# Fungsi Text – SUBSTRING\_INDEX()

---

Code Editor



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```
1 SELECT StudentID, SUBSTRING_INDEX(Email, '@', 1) as Name
2 FROM students;
```

| StudentID | Name          |
|-----------|---------------|
| 1         | Jose Mohit    |
| 2         | lala karlina  |
| 3         | Sultan Hadi   |
| 4         | jaya usman    |
| 5         | aniali wijaya |

# Fungsi Text – SUBSTR()

---

Syntax:

```
SELECT SUBSTR(columnName, Start Index, Number of string to  
be extract)  
FROM TableName;
```

Keterangan:

*columnName* --> nama kolom yang akan dicari substring-nya

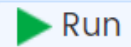
*Start Index* --> indeks dari text yang dimiliki (dimulai dari 1)

*Number of string to be extract* --> jumlah karakter atau beberapa karakter yang akan diambil.

# Fungsi Text – SUBSTR()

---

Code Editor



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```
1 SELECT StudentID, SUBSTR(FirstName, 2, 3) as Initial
2 FROM students;
```

| StudentID | Initial |
|-----------|---------|
| 1         | ose     |
| 2         | ala     |
| 3         | ult     |
| 4         | aya     |
| 5         | nja     |



## Fungsi Text – LENGTH()

Code Editor

Run

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```
1 SELECT StudentID, FirstName, LENGTH(FirstName) as Total_Char  
2 FROM students;
```

| StudentID | FirstName | Total_Char |
|-----------|-----------|------------|
| 1         | Jose      | 4          |
| 2         | Lala      | 4          |
| 3         | Sultan    | 6          |
| 4         | Jaya      | 4          |
| 5         | Anjali    | 6          |

# Fungsi Text – REPLACE()

Syntax:

```
SELECT REPLACE(ColumnName, Character/String to be change, New String/Character)  
FROM TableName;
```

Keterangan:

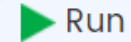
*ColumnName* --> nama kolom yang akan diganti isi tiap *record*/barisnya berdasarkan string/karakter tertentu  
*Character/String to be change* --> string/karakter yang dimiliki untuk diganti  
*New String/Character* --> string/karakter baru pengganti string/karakter sebelumnya





# Fungsi Text – REPLACE()

Code Editor



Run



Submit



0 Hint



```
1 SELECT StudentID, Email, REPLACE(Email, 'yahoo', 'gmail') as New_Email
2 FROM students;
```

| StudentID | Email                         | New_Email               |
|-----------|-------------------------------|-------------------------|
| 1         | Jose_Mohit@gmail.com          | Jose_Mohit@gmail.com    |
| 2         | <u>lala_karlina@yahoo.com</u> | lala_karlina@gmail.com  |
| 3         | Sultan_Hadi@gmail.com         | Sultan_Hadi@gmail.com   |
| 4         | jaya_usman@yahoo.com          | jaya_usman@gmail.com    |
| 5         | anjali_wijaya@yahoo.com       | anjali_wijaya@gmail.com |

# Quiz 2:

Gunakan fungsi **UPPER()** untuk mengubah kolom **FirstName** menjadi seluruhnya kapital dan gunakan **LOWER()** untuk mengubah kolom **LastName** menjadi seluruhnya non-kapital. Gunakan kedua fungsi tersebut dalam satu SELECT-Statement.

---

| StudentID | FirstName | LastName |
|-----------|-----------|----------|
| 1         | JOSE      | mohit    |
| 2         | LALA      | karlina  |
| 3         | SULTAN    | hadi     |
| 4         | JAYA      | usman    |
| 5         | ANJALI    | wijaya   |



# Fungsi Aggregate

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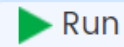
| Fungsi  | Deskripsi                                                              |
|---------|------------------------------------------------------------------------|
| SUM()   | Digunakan untuk menjumlahkan sekelompok nilai (baris) dalam satu kolom |
| COUNT() | Digunakan untuk menghitung jumlah baris                                |
| AVG()   | Digunakan untuk menghitung nilai rata - rata dari suatu kolom          |
| MIN()   | Digunakan untuk menghitung nilai minimum dari suatu kolom              |
| MAX()   | Digunakan untuk menghitung nilai maximum dari suatu kolom              |
| FIRST() | Mengembalikan nilai pada baris pertama dari suatu kolom                |
| LAST()  | Mengembalikan nilai pada baris terakhir dari suatu kolom               |



# Fungsi Aggregate – SUM()

---

Code Editor



Run



Submit



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```
1 SELECT SUM(Semester1) as Total_1, SUM(Semester2) as Total_2
2 FROM students;
```

| Total_1 | Total_2 |
|---------|---------|
| 361.7   | 356.35  |

# Fungsi Aggregate – COUNT()

---

Code Editor

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```
1 SELECT COUNT(FirstName) as Total_Student
2 FROM students;
```

| Total_Student |
|---------------|
| 5             |

# Fungsi Aggregate – AVG()

---

**Code Editor** ▶ Run ⬆ Submit 💡 0 Hint ⋮

```
1 SELECT AVG(Semester1) as AVG_1, AVG(Semester2) as AVG_2
2 FROM students;
```

| AVG_1 | AVG_2 |
|-------|-------|
| 72.34 | 71.27 |

# Quiz 3:

Gunakan fungsi **MIN()** dan **MAX()** untuk menghitung nilai dari kolom **Semester1** dan **Semester2**. Kedua fungsi tersebut dalam satu SELECT-Statement.

---

| Min1  | Max1  | Min2  | Max2  |
|-------|-------|-------|-------|
| 45.32 | 92.25 | 50.25 | 90.75 |

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# Terima Kasih